Enterprise VLAN & Trunking Lab

Project Report

1. Project Title

Enterprise VLAN and Trunking Lab Using Cisco Packet Tracer

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2. Project Overview

Modern enterprise networks must support multiple departments while maintaining security, performance, and ease of management. One of the most effective methods for achieving this is through the use of **Virtual Local Area Networks (VLANs)** and **802.1Q trunking**.

This project simulates the design and implementation of a segmented enterprise Local Area Network (LAN) using Cisco switches in Cisco Packet Tracer. The network is divided into multiple logical departments, each operating within its own VLAN. Trunk links are configured between switches to allow VLAN traffic to traverse the network while preserving VLAN separation.

The project demonstrates fundamental Layer 2 networking concepts that are essential for Network Engineers and aligns with Cisco CCNA networking objectives.

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3. Project Objectives

The primary objectives of this project are to:

- Design an enterprise Layer 2 switched network.
- Create multiple VLANs for different departments.
- Assign switch ports to the appropriate VLANs.
- Configure IEEE 802.1Q trunk links between switches.
- Implement logical network segmentation.
- Prevent communication between different VLANs.
- Verify connectivity within the same VLAN.
- Practice Cisco IOS configuration commands.
- Develop troubleshooting skills for VLAN-related issues.
- Produce professional documentation suitable for a technical portfolio.

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4. Business Scenario

A medium-sized company named **TechCorp** is expanding its network infrastructure. The organization has four departments:

- Management
- Human Resources (HR)
- Sales
- Information Technology (IT)

Each department requires its own isolated network to improve security, reduce broadcast traffic, and simplify network administration.

The IT department has been tasked with designing and implementing a VLAN-based network that separates each department while allowing future expansion through inter-VLAN routing.

At this stage of the project, communication between different VLANs is intentionally disabled. Inter-VLAN routing will be implemented in a future project.

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5. Project Scope

The project includes:

- Two Cisco Layer 2 switches
- Eight end devices (PCs)
- Multiple VLANs
- One trunk connection between switches
- Static IP addressing
- VLAN verification
- Connectivity testing
- Network documentation

The project does **not** include:

- Routers
- Inter-VLAN routing
- DHCP
- Internet connectivity
- Network Address Translation (NAT)
- Dynamic routing protocols

These features will be implemented in future projects.

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6. Network Topology

The network consists of two Cisco switches connected through an IEEE 802.1Q trunk link.

Each switch provides access ports for departmental computers.

Each department belongs to a dedicated VLAN and IP subnet.

The logical topology contains:

- Switch 1
- Switch 2
- Four VLANs
- Eight workstations
- One trunk link

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7. VLAN Design

VLAN ID	Department	Purpose
10	Management	Executive users
20	Human Resources	HR department
30	Sales	Sales department
40	Information Technology	IT department
99	Native VLAN	Native trunk VLAN

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8. IP Addressing Plan

VLAN	Network
VLAN 10	192.168.10.0/24
VLAN 20	192.168.20.0/24
VLAN 30	192.168.30.0/24
VLAN 40	192.168.40.0/24

Each workstation receives a manually configured IP address within its respective subnet.

No default gateway is configured because inter-VLAN routing is outside the scope of this project.

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9. Technologies Used

Networking

- Ethernet
- IEEE 802.1Q
- VLAN
- Trunking
- Layer 2 Switching

Software

- Cisco Packet Tracer
- Cisco IOS CLI

Documentation

- GitHub
- Draw.io
- Markdown

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10. Technical Concepts Covered

The project reinforces the following networking concepts:

- Layer 2 Switching
- MAC Address Tables
- Broadcast Domains
- Collision Domains
- VLAN Segmentation
- VLAN Tagging
- IEEE 802.1Q
- Access Ports
- Trunk Ports
- Native VLAN
- IP Addressing
- Subnetting
- Cisco IOS Configuration
- Network Verification
- Network Troubleshooting

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11. Expected Results

After configuration:

Devices within the same VLAN

- Can communicate successfully.
- Can exchange ICMP packets.
- Are located within the same broadcast domain.

Devices in different VLANs

- Cannot communicate.
- Broadcast traffic remains isolated.
- Security between departments is maintained.

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12. Verification Methods

The implementation will be validated using:

- Ping tests
- Cisco IOS verification commands
- VLAN membership verification
- Trunk verification
- MAC address table inspection

Common commands include:

- `show vlan brief`
- `show interfaces trunk`
- `show running-config`
- `show interfaces switchport`
- `show mac address-table`

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13. Expected Learning Outcomes

Upon successful completion of this project, the learner will be able to:

- Design a basic enterprise switched network.
- Configure VLANs using Cisco IOS.
- Configure access and trunk ports.

- Understand VLAN tagging using IEEE 802.1Q.
- Explain broadcast domain separation.
- Verify switch configurations.
- Troubleshoot Layer 2 connectivity problems.
- Document enterprise network configurations.
- Apply networking best practices.

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14. Portfolio Value

This project demonstrates practical experience with enterprise switching technologies commonly used in corporate environments.

Skills demonstrated include:

- VLAN implementation
- LAN segmentation
- Cisco switch configuration
- Network verification
- Technical documentation
- Basic network troubleshooting

These skills directly align with responsibilities typically expected from Junior Network Engineers, Network Support Engineers, and CCNA-certified professionals.

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15. Future Enhancements

Future versions of this project will include:

- Router-on-a-Stick (Inter-VLAN Routing)
- Dynamic Host Configuration Protocol (DHCP)
- Access Control Lists (ACLs)
- Rapid Spanning Tree Protocol (RSTP)
- EtherChannel (LACP)
- Switch Management VLAN
- SSH Remote Management
- Port Security
- Network Monitoring
- Network Automation using Python and Ansible

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16. Conclusion

This project serves as a foundational enterprise networking exercise that introduces VLAN-based network segmentation and IEEE 802.1Q trunking. By implementing and verifying multiple VLANs across interconnected Cisco switches, the project demonstrates how logical network segmentation enhances security, scalability, and manageability.

Beyond meeting CCNA-level objectives, the project emphasizes professional documentation, verification, and troubleshooting practices that mirror real-world network engineering workflows. It establishes a strong foundation for more advanced networking topics such as inter-VLAN routing, dynamic routing protocols, network security, and network automation, making it an excellent first portfolio project for aspiring Network Engineers.